



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2004

PHSC Examination

YEAR 11

Mathematics Extension

Sample Solutions

Question	Marker
1	PSP
2	Mr Choy
3	Mr Hespe
4	Mr Bigelow

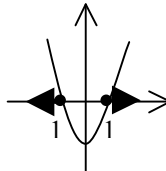
Question 1

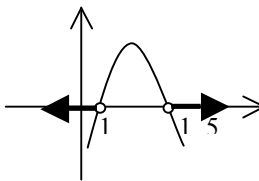
$$\begin{aligned}
 \text{(a)} \quad \frac{1}{3\sqrt{2}} + \frac{1}{3+\sqrt{2}} &= \frac{3+\sqrt{2}+3\sqrt{2}}{(3\sqrt{2})(3+\sqrt{2})} \\
 &= \frac{6}{9-2} \\
 &= \frac{6}{7}
 \end{aligned}$$

$$\frac{6}{7} = \frac{1}{3\sqrt{2}} + \frac{1}{3+\sqrt{2}} \text{ is a rational number}$$

QED

$$\begin{aligned}
 \text{(b)} \quad \text{(i)} \quad |2x-1| &= 5 \\
 2x-1 &= 5 \text{ or } 2x-1 = -5 \\
 2x &= 6, \quad 4 \\
 x &= 3, \quad 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad x^2-1 &= x^2-1-0 \\
 (x-1)(x+1) &= 0 \\
 x-1, x+1 &
 \end{aligned}$$


$$\begin{aligned}
 \text{(iii)} \quad \frac{1}{x-1} < 2 &\quad \frac{1}{x-1} - 2 < 0 \\
 \frac{1-2(x-1)}{x-1} < 0 &\quad \frac{1-2x+2}{x-1} < 0 \\
 \frac{3-2x}{x-1} < 0 &\quad (x-1)^2 \\
 (x-1)(3-2x) < 0 & \\
 x < 1, x > \frac{3}{2} &
 \end{aligned}$$


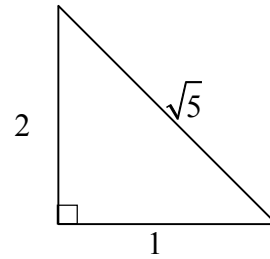
$$\text{(c)} \quad P(x) = 2x^3 - 3x^2 + x - 4$$

By the Remainder Theorem:

$$\text{Remainder} = P(2) = 16 - 12 + 2 - 4 = 2$$

$$\text{(d)} \quad \frac{x^3-1}{x^2-2x+1} = \frac{(x-1)(x^2+x+1)}{(x-1)^2} = \frac{x^2+x+1}{x-1}$$

$$\begin{aligned}
 \text{(e)} \quad \sin \frac{\pi}{4} + \frac{1}{4} &= \sin \frac{\pi}{4} \cos \frac{\pi}{4} + \cos \frac{\pi}{4} \sin \frac{\pi}{4} \\
 &= \frac{1}{\sqrt{2}} (\sin \frac{\pi}{4} + \cos \frac{\pi}{4}) \\
 &= \frac{1}{\sqrt{2}} \left(\frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}} \right) \\
 &= \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}
 \end{aligned}$$



$$\begin{aligned}
 \text{(f)} \quad y &= \frac{1}{4}(x^2 - 2x + 9) \\
 4y &= x^2 - 2x + 9 \quad x^2 - 2x + 1 + 8 \\
 (x - 1)^2 &= 4y - 8 = 4(y - 2) \\
 a &= 1
 \end{aligned}$$

Vertex $(1, 2)$, Focus $(1, 3)$

$$\begin{aligned}
 \text{(g)} \quad \text{LHS} &= \cos 3 \\
 &= \cos(2 + 1) \\
 &= \cos 2 \cos 1 - \sin 2 \sin 1 \\
 &= (2 \cos^2 1 - 1) \cos 1 - (2 \sin 1 \cos 1) \sin 1 \\
 &= 2 \cos^3 1 - \cos 1 - 2 \cos 1 \sin^2 1 \\
 &= 2 \cos^3 1 - \cos 1 - 2 \cos 1 (1 - \cos^2 1) \\
 &= 2 \cos^3 1 - \cos 1 - 2 \cos 1 + 2 \cos^3 1 \\
 &= 4 \cos^3 1 - 3 \cos 1 \\
 &= \text{RHS}
 \end{aligned}$$

QED

Question 2

Question 2

$$(a) (i) \frac{d}{dx} (1 + 2x - 4x^2 - x^3)$$

$$= [2 - 8x - 3x^2] \cdot 2$$

$$(ii) \frac{d}{dx} (1 - x^2)^{\frac{1}{2}}$$

$$= \frac{1}{2} (1 - x^2)^{-\frac{1}{2}} (-2x)$$

$$= \left[-\frac{x}{(1 - x^2)^{\frac{1}{2}}} \right] \cdot 2$$

$$(iii) \frac{d}{dx} (x-1)^4 (3x+1)$$

$$= (3x+1) 4(x-1)^3 + (x-1)^4 \cdot 3$$

$$= (x-1)^3 (12x+4+3x-3)$$

$$= [(x-1)^3 (15x+1)] \cdot 2$$

$$(iv) \frac{d}{dx} \left(\frac{2}{x^3-1} \right) = 2 \frac{d}{dx} (x^3-1)^{-1}$$

$$= -2 (x^3-1)^{-2} (3x^2)$$

$$= \left[-6x^2 (x^3-1)^{-2} \right] \cdot 2$$

$$(i) \sin x - \sqrt{3} \cos x = A \sin(x-\alpha)$$

$$A (\sin x \cos \alpha - \cos x \sin \alpha)$$

$$\therefore A \cos \alpha = 1$$

$$A \sin \alpha = \sqrt{3}$$

$$\therefore \tan \alpha = \sqrt{3}, \alpha = \frac{\pi}{3}$$

$$A^2 = 1^2 + 3 = 4, A = 2$$

$$\therefore \boxed{\sin x - \sqrt{3} \cos x = 2 \sin(x - \frac{\pi}{3})}$$

$$(ii) 2 \sin(x - \frac{\pi}{3}) = \frac{2}{\sqrt{2}}$$

$$\therefore \sin(x - \frac{\pi}{3}) = \sin \frac{\pi}{4}$$

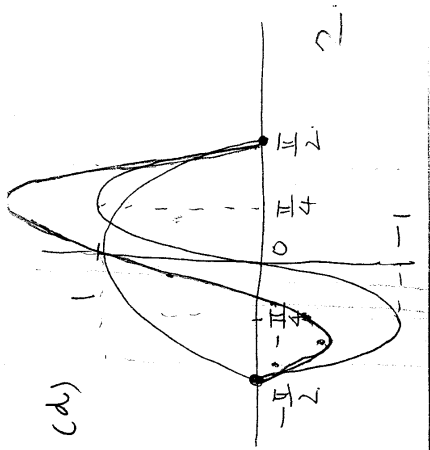
$$x - \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$\therefore \boxed{x = \left(n\pi + \frac{\pi}{3}\right) + (-1)^n \frac{\pi}{4}}$$

$$(c) (x-1) \left[(x-1) - \frac{2}{5} \right] < 0$$

$$(x-1)(x-5) < 0$$

$$\therefore \boxed{1 < x < 5}$$



$$(e) f(x) = x^3 - 19x - 30$$

$$(i) f(-2) = -8 + 38 - 30 = 0$$

$$\therefore (x+2) \text{ is a factor}$$

$$x+2 \overline{) x^3 - 19x - 30}$$

$$\underline{-(x^3 + 2x^2)}$$

$$\underline{-2x^2 - 19x - 30}$$

$$\underline{-(-2x^2 - 4x)}$$

$$\underline{-15x - 30}$$

$$\underline{-(-15x - 30)}$$

$$\underline{0}$$

$$\therefore (ii) \boxed{P(x) = (x+2)(x+3)(x-5)}$$

Question 3

3. (a) (i) Express the decimal $0.1\dot{5}\dot{4}$ as a common fraction in lowest terms.

Solution:

$$\begin{aligned}x &= 0.1\dot{5}\dot{4}, \\100x &= 15.4\dot{5}\dot{4}, \\99x &= 15.3, \\x &= \frac{153}{990}, \\&= \frac{17}{110}.\end{aligned}$$

- (ii) Find $\log_2 74$ correct to three decimal places.

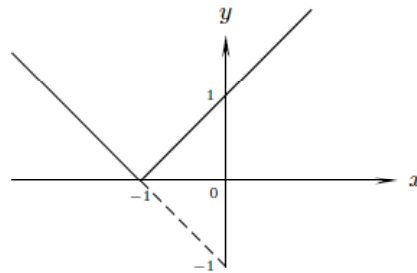
Solution:

$$\begin{aligned}\log_2 74 &= \frac{\log 74}{\log 2}, \\&\approx 6.209 \text{ [6.20945336562 on calculator]}.\end{aligned}$$

- (b) Draw neat sketches of the following functions, showing their principal features:

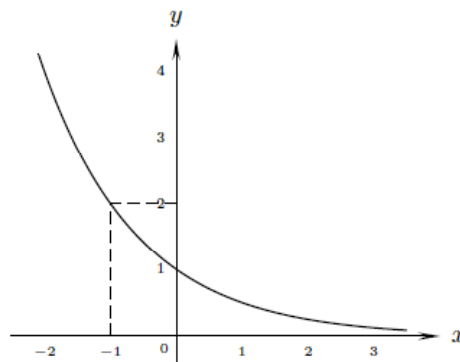
- (i) $y = |x + 1|$

Solution: When $x < -1$, $y = -x - 1$; when $y \geq -1$, $y = x + 1$.



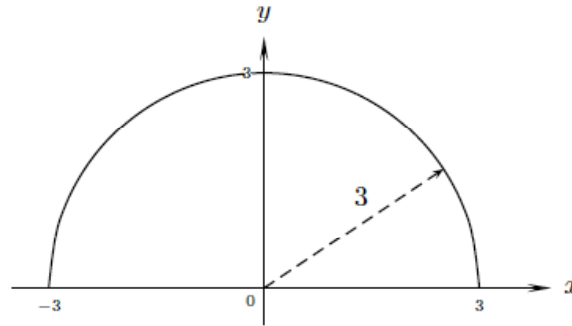
- (ii) $y = 2^{-x}$

Solution:



(iii) $y = \sqrt{9 - x^2}$

Solution:



(c) Given the function $f(x) = \frac{x}{x^2 + 1}$

(i) Find $f(-1)$

Solution:
$$f(-1) = \frac{-1}{(-1)^2 + 1},$$
$$= -\frac{1}{2}.$$

(ii) Show that $f(x)$ is odd.

Solution:
$$f(-x) = \frac{-x}{(-x)^2 + 1},$$
$$= -\frac{x}{x^2 + 1},$$
$$= -f(x).$$

$\therefore f(x)$ is odd.

(iii) Find x such that $f(x) = 0$.

Solution:
$$\frac{x}{x^2 + 1} = 0,$$
$$\therefore x = 0.$$

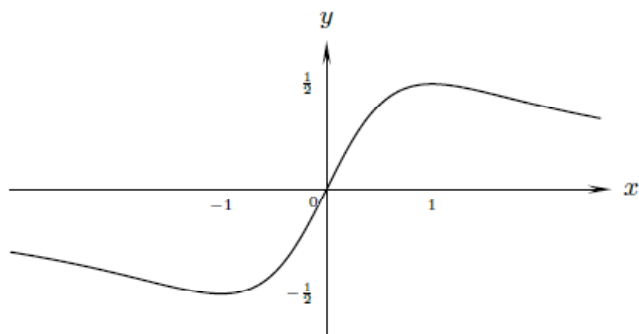
(iv) State the domain and range of $f(x)$.

Solution: Domain: $x \in \mathbb{R}$, or all real x .
Now, putting $y = f(x)$ and rearranging,
$$yx^2 + y = x,$$
$$yx^2 - x + y = 0,$$
$$\Delta = 1 - 4y^2 \geq 0 \text{ for real values of } x,$$
$$\text{i.e. } \frac{1}{4} = y^2.$$

And thus the range is $-\frac{1}{2} \leq f(x) \leq \frac{1}{2}$.

(v) Sketch the function.

Solution:



(d) Of the three roots of the cubic equation $x^3 - 15x + 4 = 0$, two are reciprocals.

(i) Find the other root.

Solution: Let the roots be α , $\frac{1}{\alpha}$, β , then

$$\alpha \times \frac{1}{\alpha} \times \beta = -4 \text{ (product of roots),}$$

$$\text{i.e. } \beta = -4.$$

(ii) Find the reciprocal roots.

Solution: $\alpha + \frac{1}{\alpha} - 4 = 0$ (sum of roots),

$$\alpha^2 - 4\alpha + 1 = 0,$$

$$\alpha = \frac{4 \pm \sqrt{16 - 4}}{2},$$

$$= 2 \pm \sqrt{3}.$$

i.e. the reciprocal roots are $2 \pm \sqrt{3}$.

(e) Find the distance between the parallel lines $4x + 3y = 12$ and $4x + 3y = 5$.

Solution: One point on $4x + 3y = 12$ is $(0, 4)$.

$$\therefore \text{ Distance} = \frac{|0 \times 4 + 4 \times 3 - 5|}{\sqrt{16 + 9}},$$

$$= \frac{7}{5}.$$

Question 4

$$(2) \quad (i) \quad \frac{AM}{c} = \cos A \quad \left| \quad \frac{AN}{b} = \cos A. \right.$$

$$\therefore \underline{AM = c \cos A.} \quad \left| \quad \therefore \underline{AN = b \cos A.} \right.$$

$$(ii) \quad MN^2 = AN^2 + AM^2 - 2AN \cdot AM \cdot \cos A.$$

$$= b^2 \cos^2 A + c^2 \cos^2 A - 2bc \cos^2 A \cdot \cos A.$$

$$= \cos^2 A (b^2 + c^2 - 2bc \cos A) \quad \left(\begin{array}{l} \text{NB.} \\ a^2 = b^2 + c^2 - 2bc \cos A \\ \text{from Cosine Rule.} \end{array} \right)$$

$$= \cos^2 A \cdot a^2$$

$$\therefore \underline{MN = a \cos A.}$$

$$(b) \quad (i) \quad y - px + ap^r = 0.$$

$$(ii) \quad m_{sp} = \frac{ap^r - a}{2ap}$$

$$= \frac{a(p^r - 1)}{2ap}$$

$$= \frac{p^r - 1}{2p}$$

$$\tan \theta = \left| \frac{\frac{p^r - 1}{2p} - p}{1 + \frac{p^r - 1}{2p} \times p} \right|$$

$$= \left| \frac{\frac{p^r - 1 - 2p^r}{2p}}{\frac{2 + p^r - 1}{2}} \right|$$

$$= \left| \frac{-(1 + p^r)}{p(1 + p^r)} \right|$$

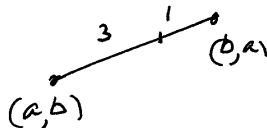
$$= \left| \frac{-1}{p} \right|$$

$$= \frac{1}{|p|}$$

(iii) $\tan \theta$ is undefined when $p = 0$. At this point $(0, 0)$ the angle is 90° .

$$\begin{aligned}
 \text{(c)} \quad \underline{\text{LHS}} &= \cot \theta + \tan \theta \\
 &= \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \\
 &= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} \\
 &= \frac{1}{\sin \theta \cos \theta} \\
 &= \frac{2}{2 \sin \theta \cos \theta} \\
 &= \frac{2}{\sin 2\theta} \\
 &= 2 \csc 2\theta \\
 &= \underline{\text{RHS.}}
 \end{aligned}$$

(d)



$$\frac{3b+a}{4} = 0 \Rightarrow 3b+a=0 \quad \text{--- (1)}$$

$$\frac{3a+b}{4} = 4 \Rightarrow 3a+b=16 \quad \text{--- (2)}$$

Adding (1) + (2)
simultaneously.
From (1) $a = -3b$.
Substitute in (2)

$$-9b+b=16$$

$$-8b=16$$

$$\underline{b = -2.}$$

Sub in (1)

$$-6+a=0$$

$$\underline{a = 6.}$$

$$\therefore \underline{a=6, b=-2.}$$

$$\text{(e)} \quad (1) \quad \frac{a}{AP} = \tan \theta \therefore AP = \frac{a}{\tan \theta} \quad \text{Also} \quad \frac{b}{BP} = \tan \theta \therefore BP = \frac{b}{\tan \theta}.$$

By Pythagoras $d^2 = AP^2 + BP^2$

$$d^2 = \frac{a^2}{\tan^2 \theta} + \frac{b^2}{\tan^2 \theta} \quad \text{--- (A)}$$

$$\begin{aligned}
 \text{(1)} \quad \underline{\text{LHS}} &= \tan^2 \alpha + \tan^2 \beta = \frac{a^2}{d^2} + \frac{b^2}{d^2} \\
 &= \frac{a^2 + b^2}{d^2} \\
 &= \frac{d^2 \tan^2 \theta}{d^2} \\
 &= \tan^2 \theta \\
 &= \underline{\text{RHS.}}
 \end{aligned}$$

(From (A))
 $a^2 + b^2 = d^2 \tan^2 \theta.$



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2004

YEAR 11

**PRELIMINARY HIGH SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – One and a half hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.

Total Marks – 72

- Attempt all questions.
- All questions are of equal value.
- Each question is to be answered in a separate booklet.

Examiner: *A.M.Gainford*

Question 1. (18 Marks)

- (a) Show that $\frac{1}{3-\sqrt{2}} + \frac{1}{3+\sqrt{2}}$ is a rational number. 2
- (b) Solve for x : 6
- (i) $|2x-1|=5$
- (ii) $x^2 \geq 1$
- (iii) $\frac{1}{x-1} < 2$
- (c) Find the remainder when the polynomial $P(x) = 2x^3 - 3x^2 + x - 4$ is divided by $x - 2$. 1
- (d) Simplify $\frac{x^3-1}{x^2-2x+1}$. 2
- (e) If $\tan \theta = 2$, and $0 < \theta < \frac{\pi}{2}$, find the exact value of $\sin\left(\theta + \frac{\pi}{4}\right)$. 2
- (f) Find the vertex and focus of the parabola $y = \frac{1}{4}(x^2 - 2x + 9)$. 2
- (g) Show that for all θ : 3
- $$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta.$$

Question 2. (18 Marks)

- (a) Differentiate: 8
- (i) $1 + 2x - 4x^2 - x^3$
- (ii) $\sqrt{1-x^2}$
- (iii) $(x-1)^4(3x+1)$
- (iv) $\frac{2}{x^3-1}$
- (b) (i) Express $\sin x - \sqrt{3}\cos x$ in the form $A\sin(x - \alpha)$, where $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Find the general solution to the equation $\sin x - \sqrt{3}\cos x = \frac{2}{\sqrt{2}}$. 2

- (c) Solve $(x-1)^2 < 4(x-1)$, and graph the solution on the number line. 2
- (d) Sketch the graph of $y = \cos x + \sin 2x$ in the domain $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. 2
- (e) Given the polynomial $P(x) = x^3 - 19x - 30$. 2
- (i) Use the factor theorem to find a zero of the polynomial.
- (ii) Express $P(x)$ as a product of three linear factors.

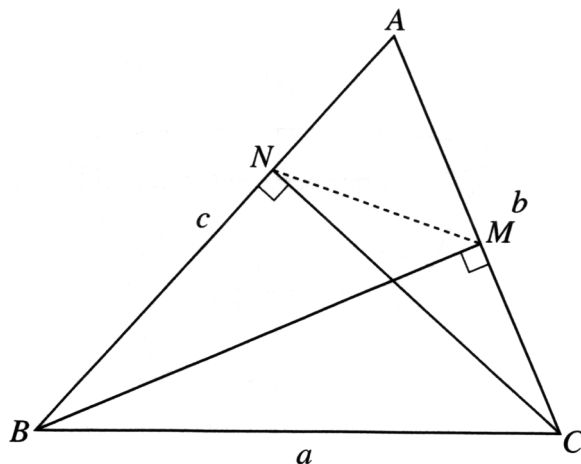
Question 3. (18 Marks)

- (a) (i) Express the decimal $0.1\dot{5}\dot{4}$ as a common fraction in lowest terms. 2
- (ii) Find $\log_2 74$ correct to three decimal places.
- (b) Draw neat sketches of the following functions, showing their principle features: 6
- (i) $y = |x + 1|$ (ii) $y = 2^{-x}$ (iii) $y = \sqrt{9 - x^2}$
- (c) Given the function $f(x) = \frac{x}{x^2 + 1}$ 6
- (i) Find $f(-1)$.
- (ii) Show that $f(x)$ is odd.
- (iii) Find x such that $f(x) = 0$.
- (iv) State the domain and range of $f(x)$.
- (v) Sketch the function.
- (d) Of the three roots of the cubic equation $x^3 - 15x + 4 = 0$, two are reciprocals. 2
- (i) Find the other root.
- (ii) Find the reciprocal roots.
- (e) Find the distance between the parallel lines $4x + 3y = 12$ and $4x + 3y = 5$. 2

Question 4. (18 Marks)

(a)

4



Triangle ABC has sides of length a, b, c as shown.
 BM is perpendicular to AC and CN is perpendicular to AB .

- (i) Show that $AM = c \cos A$ and $AN = b \cos A$.
- (ii) Hence, using the cosine rule, prove that $MN = a \cos A$.

(b) Let $P(2ap, ap^2)$ be a point on the parabola $x^2 = 4ay$.

4

- (i) Write down the equation of the tangent at P .
- (ii) Let θ be the acute angle between the tangent at P and the line SP , which joins P with the focus S .

Show that $\tan \theta = \frac{1}{|p|}$.

- (iii) Explain the situation at the one point where this angle is not acute.

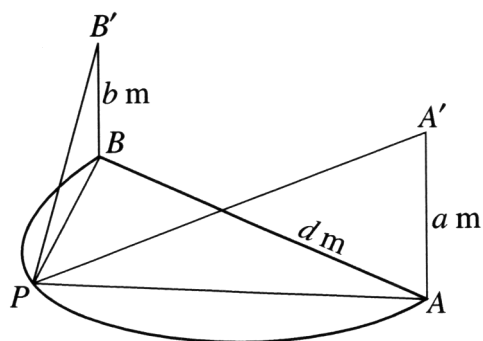
(c) Show that $\cot \theta + \tan \theta = 2 \operatorname{cosec} 2\theta$.

2

(d) The point $P(0, 4)$ divides the interval from (a, b) to (b, a) in ratio 3 : 1.
 Find the values of a and b .

2

- (e) APB is a horizontal semicircle, diameter d m.
 At A and B are vertical posts of height a m and b m respectively.
 From P , the angle of elevation of the tops of both posts is θ .
 The angle APB is a right angle.



- (i) Prove that $d^2 = \frac{a^2}{\tan^2 \theta} + \frac{b^2}{\tan^2 \theta}$.
- (ii) From B , the angle of elevation of A' is α , and from A , the angle of elevation of B' is β .

Prove that $\tan^2 \alpha + \tan^2 \beta = \tan^2 \theta$.

End of the paper.